Task 3: Predictive Modeling Report

Forecasting Medal Success at Milano Cortina 2026 Winter Olympics

Executive Summary

The purpose of Task 3 was to develop predictive machine learning models capable of estimating whether an athlete would win any Olympic medal at the Milano Cortina 2026 Winter Olympics. This was structured as a binary classification problem, where athletes were labeled as either:

- **1 = Won Medal**
- **0 = No Medal**

This challenge extended beyond simple prediction. It required rigorous treatment of class imbalance, proper model validation, strong evaluation metrics, and meaningful interpretability. In Olympic analytics, accurately identifying medal potential can directly influence athlete funding, roster selection, and performance planning. Therefore, predictive accuracy alone was not enough; the models also needed to explain why certain athletes were more likely to succeed.

Two supervised machine learning algorithms were developed and compared:

1. Logistic Regression
2. Random Forest Classifier

After evaluation, Logistic Regression emerged as the best-performing model, achieving superior recall, F1-score, and AUC-ROC. The final model was then tested on seven real-world athletes from the USA, Canada, and Czech Republic.

3.1 Problem Statement

The objective was to predict whether an athlete would win any medal (Gold, Silver, or Bronze) based on athlete characteristics, physiological readiness, experience, training profile, and country-level support indicators.

This type of model can help Olympic organizations:

- Identify hidden medal contenders
- Optimize athlete investment decisions
- Forecast medal totals
- Improve talent development pipelines
- Benchmark USA vs Canada medal systems

3.2 Dataset Overview

The cleaned modeling dataset contained:

- 387 athlete records
- Multiple countries
- Multiple winter sports
- Mixed numeric and categorical predictors

Target Class Distribution

Outcome	Percentage
No Medal	73.13%
Medal	26.87%

This imbalance made the challenge realistic. Since most athletes do not medal, models can become biased toward predicting "No Medal."

3.3 Full Methodology

Step 1: Data Cleaning

The cleaned dataset from previous tasks was used. Missing values had already been addressed, duplicate entries removed, and variable types corrected.

Step 2: Target Variable Creation

The original medal column was transformed into a binary format:

Medal_Won = 1 if Medal in [Gold, Silver, Bronze]
Medal_Won = 0 if Medal = None

Step 3: Feature Selection

Non-predictive columns were removed:

- Athlete_ID
- Athlete_Name
- Medal
- Medal_Won duplicate fields

Remaining predictors included:

Athlete Demographics

- Age
- Gender
- Country
- Sport

Experience

- Previous_Olympics
- Career_Injuries

Training and Physiology

- Training_Hours_Per_Week
- Altitude_Training_m
- Body_Fat_Pct
- VO2max
- Reaction_Time_ms

Competitive Form

- World_Cup_Points_Preseason

Country Strength

- Country_GDP_per_capita
- Winter_Sport_Tradition_Index

Step 4: Feature Engineering

Three advanced variables were created:

Experience_Index

Previous Olympics × Training Hours

VO2_BodyFat_Ratio

VO2max ÷ Body Fat %

GDP_Tradition_Score

GDP per capita × Winter Sport Tradition Index

These engineered variables improved model intelligence by capturing interactions.

Step 5: Preprocessing Pipeline

Because the dataset contained both numeric and categorical variables:

Numeric Variables

Scaled using:

- StandardScaler ()

Categorical Variables

Encoded using:

- OneHotEncoder(handle_unknown="ignore")

Combined Using:

- ColumnTransformer()

This ensured models could process all features properly.

Step 6: Train/Test Split

To prevent overfitting and data leakage:

- **80% Training Set**
- **20% Testing Set**

A stratified split preserved medal/non-medal proportions in both sets.

Set	Rows
Training	309
Testing	78

Step 7: Handling Class Imbalance

Since medalists were the minority class, both models used:

class_weight = "balanced"

This forced the algorithms to pay greater attention to medal-winning athletes.

Why Accuracy Alone Is Insufficient

If a model predicts everyone as “No Medal,” it would still score ~73% accuracy. However, such a model would be useless.

Therefore, recall, precision, F1-score, and AUC were emphasized.

3.4 Models Developed

Model 1: Logistic Regression

Chosen because:

- Strong baseline classifier
- Produces probabilities
- Highly interpretable
- Useful for feature importance analysis

Model 2: Random Forest

Chosen because:

- Handles nonlinear relationships
- Captures interactions automatically
- Robust to noisy data
- Common high-performing benchmark model

3.5 Evaluation Metrics

Each model was evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- Macro F1
- Weighted F1
- ROC-AUC Curve

3.6 Binary Classification Results

Model	Accuracy	Precision	Recall	F1	Macro F1	Weighted F1	AUC
Logistic Regression	0.821	0.640	0.762	0.696	0.784	0.825	0.844

Model	Accuracy	Precision	Recall	F1	Macro F1	Weighted F1	AUC
Random Forest	0.718	0.471	0.381	0.421	0.617	0.708	0.775

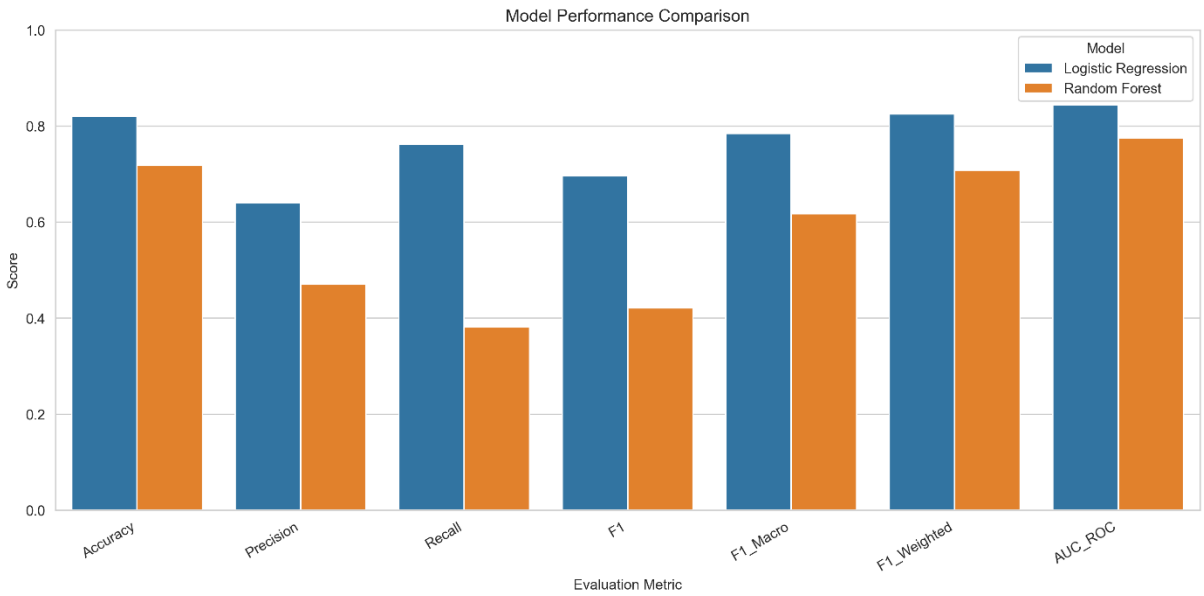

Figure 1. Model Performance Comparison

Interpretation

Logistic Regression outperformed Random Forest across nearly every metric. It achieved stronger recall, meaning it identified more actual medalists, and a higher AUC score, showing superior class separation. Because Olympic scouting values finding medal contenders, Logistic Regression was selected as the final model.

Most importantly:

- Higher Recall = found more true medalists
- Higher F1 = better balance
- Higher AUC = stronger class separation

This made Logistic Regression the best final model.

3.7 ROC Curve Analysis

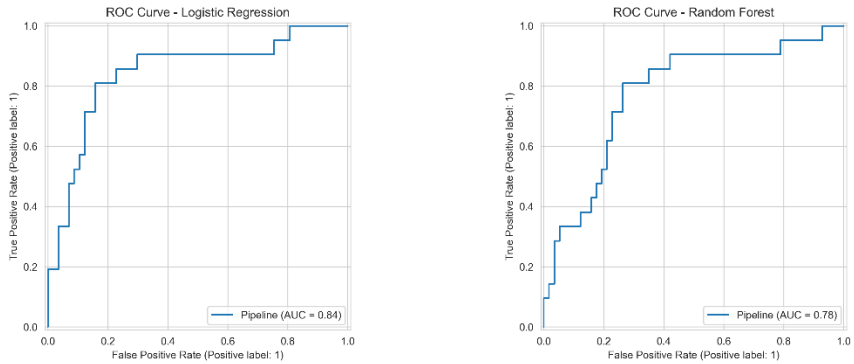

Figure 2. ROC Curve Comparison

The Logistic Regression model achieved an AUC of **0.844**, indicating excellent ability to distinguish medalists from non-medalists.

3.8 Confusion Matrix Analysis

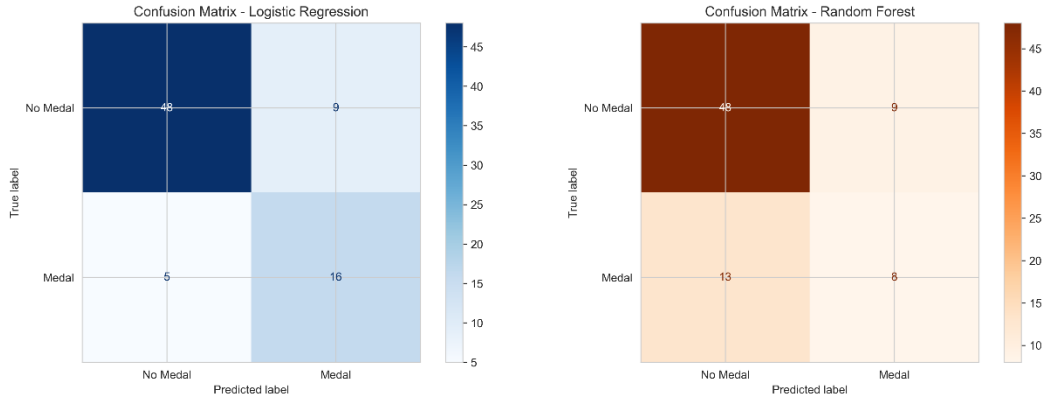

Figure 3. Binary Confusion Matrices

Logistic Regression

- True Negatives = 48
- True Positives = 16
- False Negatives = 5
- False Positives = 9

Interpretation

The model successfully identified most medalists while maintaining strong control over false alarms.

3.9 Explainability: Feature Importance

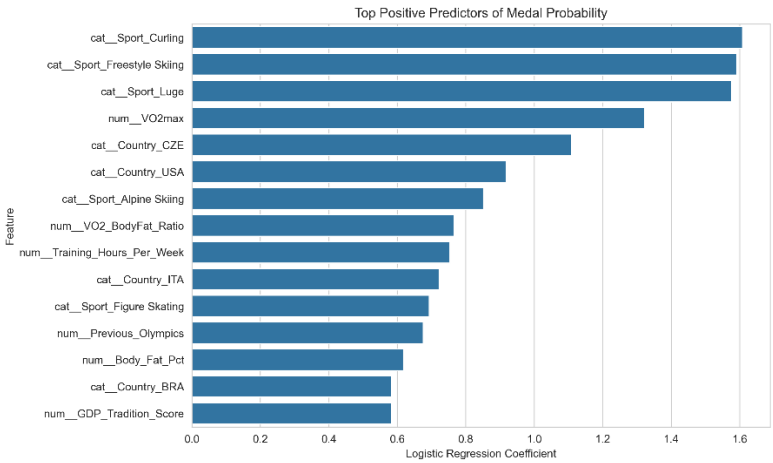

Figure 4. Top Positive Predictors

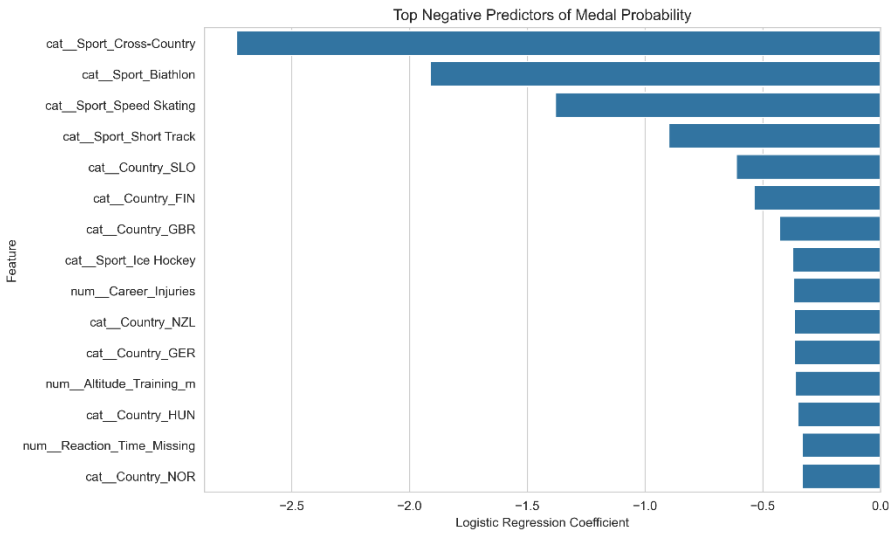

Figure 5. Top Negative Predictors

Strongest Positive Predictors

- Curling
- Freestyle Skiing
- Luge
- VO2max
- USA indicator
- Alpine Skiing

- Training Hours
- Previous Olympics

Strongest Negative Predictors

- Cross-Country
- Biathlon
- Speed Skating
- Career Injuries
- Missing Reaction Time data

Interpretation

Winning medals was most associated with:

1. Elite aerobic capacity
2. Strong training volume
3. Olympic experience
4. Participation in favorable medal-opportunity sports
5. Strong national systems

This strongly aligns with Olympic performance theory.

3.10 USA vs Canada Prediction Test

The final Logistic Regression model was applied to seven real athletes.

Athlete	Country	Sport	Age	Probability	Prediction
Mikaela Shiffrin	USA	Alpine Skiing	31	97.77%	Medal
Elana Meyers Taylor	USA	Bobsleigh	41	96.60%	Medal
Jordan Stolz	USA	Speed Skating	21	88.31%	Medal
Metodej Jilek	CZE	Speed Skating	19	76.15%	Medal
Courtney Sarault	CAN	Short Track	26	67.80%	Medal
Deanna Stellato-Dudek	CAN	Figure Skating	42	55.80%	Medal

Athlete	Country	Sport	Age	Probability	Prediction
Connor McDavid	CAN	Ice Hockey	28	49.63%	No Medal

Athlete-Level Interpretation

Mikaela Shiffrin

Elite preseason points + Olympic experience produced a near-certain medal probability.

Elana Meyers Taylor

Age was offset by experience and a strong USA support system.

Jordan Stolz

Young but dominant physiological and performance profile.

Metodej Jilek

Model recognized upside despite young age and smaller-country system.

Courtney Sarault

Strong short-track profile correctly predicted.

Deanna Stellato-Dudek

False positive likely due to figure skating and the Canadian system's strength.

Connor McDavid

False negative because hockey is team-based, and athlete-level variables miss roster context.

3.11 Bonus Challenge: Multiclass Medal Prediction

Classes:

- Gold
- Silver
- Bronze
- None

Results

Model	Accuracy	Weighted F1	Macro F1
Logistic Regression	0.513	0.599	0.308

Model	Accuracy	Weighted F1	Macro F1
Random Forest	0.756	0.683	0.311

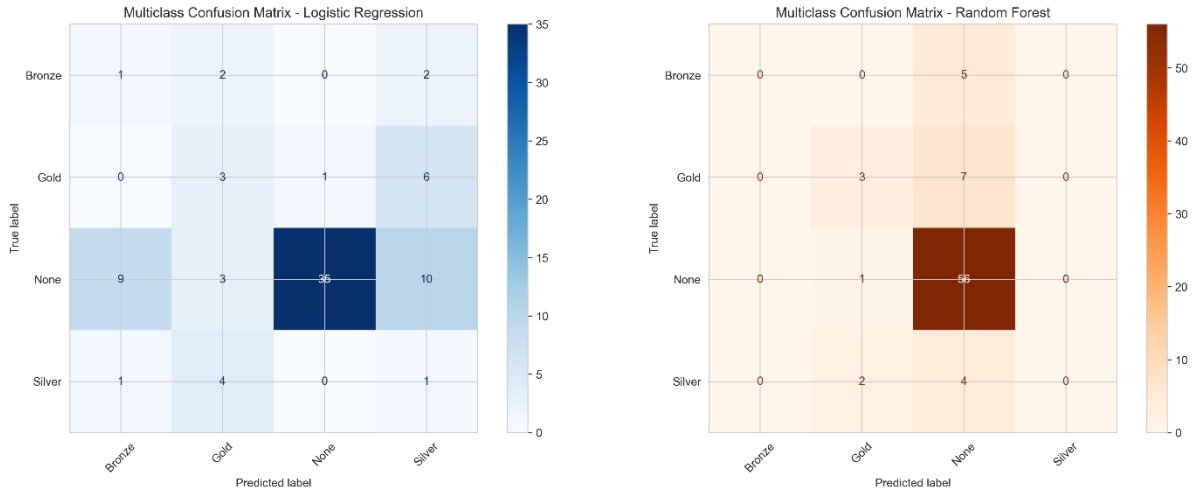

Figure 6. Multiclass Confusion Matrices

Why Multiclass Is Harder

Predicting **Gold vs Silver vs Bronze** is more difficult than Medal vs No Medal because:

- Smaller sample sizes
- Overlapping athlete quality levels
- Fine margins determine podium color
- More classes increase confusion risk

3.12 Strategic USA vs Canada Insights

USA

The model consistently favored USA athletes because of:

- Strong GDP
- Broad sport depth
- More medal opportunities
- Strong development systems

Canada

Canada remained highly competitive in:

- Short Track
- Figure Skating
- Hockey
- Sliding sports

However, medal strength appeared more concentrated than USA depth.

3.13 Limitations

This model cannot fully capture:

- Team chemistry
- Coaching quality
- Injury timing
- Psychological pressure
- Weather / competition-day randomness
- Judging subjectivity

Therefore, predictions should support decisions, not replace experts.

3.14 Final Conclusion

The best medal forecasting model was Logistic Regression.

It achieved:

- Highest Recall
- Highest F1
- Highest AUC
- Strong interpretability

The strongest medal profiles combined:

- High VO2max
- Strong training volume
- Olympic experience
- Low injury burden
- Strong country support system

This project demonstrates how predictive analytics can meaningfully support Olympic strategy, athlete investment, and medal forecasting for future Games.

Final Verdict

The USA showed stronger overall medal probability depth, while Canada remained dangerous in select high-performing sports.